

Graphwar floor construction

The article focus on how to construct approximations of floor in Graphwar. The reason of just consider approximations instead of 0 error is that there are some breakpoints of floor that Graphwar can't handle on $x \in \mathbb{Z}$.

In fact, it contains 2 subquestions:

1. How to construct a floor approximation only with operators in Graphwar without consider if it can run in Graphwar
2. How to further process of the function in **1** (if it is needed) such that the final function can run in Graphwar normally

Prerequisites

Graphwar allows and only allows the following operators:

□ + □

□ − □

□ × □

$\frac{\square}{\square}$

$\square^{\square}$

$\sqrt{\square}$

ln □

log □

|□|

sin □

cos □

tan □

exp □

Their syntaxes are:

```
1  ()+()  
2  ()-()  
3  ()*()  
4  ()/()  
5  ()^()  
6  sqrt()  
7  ln()  
8  log()  
9  abs()  
10 sin() or sen()  
11 cos()  
12 tan() or tg()  
13 exp()
```

Trivial solutions

Notice that Fourier series can be used to approximate floor:

$$\lfloor x \rfloor \approx x - \frac{1}{2} + \frac{1}{\pi} \sum_{n=1}^N \frac{\sin(2n\pi x)}{n}$$

Or, we can approximate it with the sum of several sigmoid (step) functions. Since the x -axis range of Graphwar is [-25,25)\$, we get the following construction:

$$\lfloor x \rfloor \approx -26 + \sum_{n=-25}^{25} s(x - n)$$

where

$$s(x) = \frac{1}{1 + e^{-Mx}}$$

and M is a large number

Both of them have huge problems although they can also solve the 2nd subquestion naturally.

The Fourier approximation converges so slowly that the precision is still very low after accumulating dozens of terms: the maximum error at points far from integers is about 0.02 if $N = 60$. Moreover, regardless of N as big as, the Gibbs phenomenon exists near integer points, that is, the function will overshoot and oscillate, whose error can't be eliminated.

Indeed the sum of Sigmoid functions construction can yield a high precision result, but its expression is too long (and so does Fourier form) to be used in actual situation.

The answers from Finnley

After initial attempts failed, I sent this question to the official discord server of Graphwar to ask for help. One of the member of the server, Finnley gave two pretty nice answers:

$$\lfloor x \rfloor \approx x - \left(\frac{\left(\sin\left(\frac{\pi x}{2}\right)\right)^2}{1 + e^{-500 \sin(\pi x)}} + \frac{\left(\sin\left(\frac{\pi(x-1)}{2}\right)\right)^2}{1 + e^{500 \sin(\pi x)}} + 0.1 \sin(2\pi x) \left(1 + 0.077 \sin\left(\frac{3\pi x}{\sin(\pi x)}\right)\right) \right)$$

and:

$$\lfloor x \rfloor \approx x - \frac{1}{2} + \frac{2 - \frac{4}{-1.9 \tan\left(\frac{\pi\left(x-\frac{1}{2}\right)}{2}\right)}}{3 + e^{-500 \cos(\pi\left(x-\frac{1}{2}\right))}} + \frac{2 - \frac{4}{-1.9 \tan\left(\frac{\pi\left(x-\frac{3}{2}\right)}{2}\right)}}{3 + e^{500 \cos(\pi\left(x-\frac{1}{2}\right))}}$$

They are shorter then trivial solutions while maintaining high precision, where the corresponding maximum error at the points far from integers is about 0.02 and 0.0005 and also solves the 2nd subquestion naturally.

Arctangent

My further attempts took a different path, consider $\arctan(\cot(x))$, where $\cot(x) = \frac{1}{\tan(x)}$, observe that it's a periodic function that satisfies the following relation:

$$\arctan(\cot(x)) = -x + \frac{\pi}{2} + k\pi, x \in (k\pi, (k+1)\pi), k \in \mathbb{Z}$$

After simple transformation, we get:

$$\lfloor x \rfloor = x - \frac{1}{2} + \frac{1}{\pi} \arctan(\cot(\pi x)), x \notin \mathbb{Z}$$

Approximation

It seems that the 1st subquestion have been solved by the expression, but it's a pity that there are no inverse trigonometric functions in Graphwar. So we have to construct the approximation of $\arctan$.

1st try

My first thought was the the identity: $\arctan(x) = \arcsin\left(\frac{x}{\sqrt{1+x^2}}\right)$

Approximating $\arcsin$ with $\frac{\pi}{2}x$, we get $\arctan(x) \approx \frac{\pi}{2} \frac{x}{\sqrt{1+x^2}}$

Adding an undetermined constant term c to the denominator of $\frac{x}{\sqrt{1+x^2}}$ gives $\frac{x}{\sqrt{1+x^2+c}}$

Let $f(x) = \frac{x}{\sqrt{1+x^2+c}}$, considering that $\arctan'(0) = 0$, let $f'(0) = 0$, we get $c = \frac{\pi}{2} - 1$

So

$$\arctan(x) \approx \frac{\pi}{2} \frac{x}{\sqrt{1+x^2+\frac{\pi}{2}-1}}$$

The maximum error is 0.013

So

$$\lfloor x \rfloor \approx x - \frac{1}{2} + \frac{1}{2} \frac{\cot(\pi x)}{\sqrt{1+\cot(\pi x)^2+\frac{\pi}{2}-1}}$$

The maximum error at points far from integers is about 0.004

2nd try

When I lost in the identity $\arctan(x) = \arcsin\left(\frac{x}{\sqrt{1+x^2}}\right)$ and try to get a further $\arcsin(x)$ approximation, Finnley found a better $\arctan(x)$ approximation:

Let $f(x) = \frac{ax}{b+\sqrt{c+x^2}}$

Notice that $\arctan'(0) = 1$, $\arctan(1) = \frac{\pi}{4}$, $\lim_{x \rightarrow \infty} \arctan(x) = \frac{\pi}{2}$

Solve the equation system obtained by replace $\arctan(x)$ with $f(x)$, we get:

$$\begin{cases} a = \frac{\pi}{2} \\ b = \frac{12-\pi^2}{4(4-\pi)} \approx 0.620 \\ c = \frac{(6-\pi)^2(2-\pi)^2}{16(4-\pi)^2} \approx 0.903 \end{cases}$$

So we have:

$$\arctan(x) \approx \frac{\pi}{2} \frac{x}{\sqrt{0.903 + x^2} + 0.62}$$

The maximum error is 0.002

So

$$\lfloor x \rfloor \approx x - \frac{1}{2} + \frac{1}{2} \frac{\cot(\pi x)}{\sqrt{0.903 + \cot(\pi x)^2} + 0.62}$$

The maximum error at points far from integers is about 0.0007

3rd try

Finally I found the approximation in [Inigo Quilez](#)'s website:

$$\arctan(x) \approx \frac{\pi^2 x}{4 + \sqrt{34 + (2\pi x)^2}}$$

The maximum error is 0.001525

Finally we get:

$$\lfloor x \rfloor \approx x - \frac{1}{2} + \frac{\pi \cot(\pi x)}{4 + \sqrt{34 + (2\pi \cot(\pi x))^2}}$$

The maximum error at points far from integers is about 0.000485

I chose the 3rd approximation as the final result

Eliminate breakpoints

The 1st subquestion has been solved, but the 2nd one haven't. The above expression will be terminated when it passes through the integers because their slope is too high to be handled by Graphwar or any discontinuous point is used as the sampling point.

Continuous cotangent

Consider the following identity:

$$\cot(x) = \frac{\sin(2x)}{1 - \cos(2x)}$$

Add a very small positive number to the denominator of it, we get:

$$\cot(x) \approx \frac{\sin(2x)}{1 - \cos(2x) + \varepsilon}$$

Replace $\cot(x)$ with it in the above formula in **3rd try**, by controlling the size of ε , at the cost of some precision, we can control the function's slope while eliminating breakpoints to ensure that the function does not terminate in Graphwar.

$$\lfloor x \rfloor \approx x - \frac{1}{2} + \frac{\pi \frac{\sin(2\pi x)}{1 - \cos(2\pi x) + \varepsilon}}{4 + \sqrt{34 + (2\pi \frac{\sin(2\pi x)}{1 - \cos(2\pi x) + \varepsilon})^2}}$$

In Graphwar II, because of the algorithm improvements, ε can be arbitrarily close to 0. But in Graphwar I can't do that because of the algorithm defects. The smallest value of ε is about 0.0007 if you just input a floor function. But if you want to compose it with any other function, the value should be analyzed specifically.

Finally it's the expression of floor that can be used in Graphwar: `x-0.5+(pisin(2pix)/(1-cos(2pix)+0.0007))/(4+sqrt(34+(2*pisin(2pix)/(1-cos(2pix)+0.0007))^2))`

The End?

It's hard to control the value of ε to ensure the above formula maintains high precision and does not terminate early sometime if it's used to combine with other functions in Graphwar I. Furthermore, x appears so many times that if you want to construct a composite function like $\lfloor \sin(x) \rfloor$, you have to replace x for 5 times. Is there a new way to get a better floor approximation?

Consider the following relationship between arcsin and arctan:

$$\arcsin(x) = \arctan\left(\frac{x}{\sqrt{1-x^2}}\right)$$

Replace x with $\frac{x}{\sqrt{1-x^2}}$ in the arctan approximation in **3rd try**, we get a arcsin approximation with 0.001525 maximum error:

$$\arcsin(x) \approx \frac{\pi^2 \frac{x}{\sqrt{1-x^2}}}{4 + \sqrt{34 + (2\pi \frac{x}{\sqrt{1-x^2}})^2}}$$

Similarly, the rest inverse trigonometric function approximations can be obtained in the same way.

After getting the construction, just for fun, I tried constructing an interesting expression with it in Graphwar:

$$\frac{\pi^2 \frac{\sin(x)}{\sqrt{1-\sin^2(x)}}}{4 + \sqrt{34 + (2\pi \frac{\sin(x)}{\sqrt{1-\sin^2(x)}})^2}} \approx \arcsin(\sin(x)) = \begin{cases} x - 2k\pi, & x \in [-\frac{\pi}{2} + 2k\pi, \frac{\pi}{2} + 2k\pi] \\ -x + 2k\pi, & x \in [\frac{\pi}{2} + 2k\pi, \frac{3\pi}{2} + 2k\pi] \end{cases}$$

$$k \in \mathbb{Z}$$

Finnley realized immediately that it can also be used to make another floor approximation

Consider the similar form and let it be $\phi(x)$:

$$\phi(x) = \frac{\pi^2 \frac{\cos(x)}{\sqrt{1-\cos^2(x)}}}{4 + \sqrt{34 + (2\pi \frac{\cos(x)}{\sqrt{1-\cos^2(x)}})^2}} \approx \arcsin(\cos(x)) = \begin{cases} x + \frac{\pi}{2} - 2k\pi, & x \in [-\pi + 2k\pi, 2k\pi] \\ -x + \frac{\pi}{2} + 2k\pi, & x \in [2k\pi, \pi + 2k\pi] \end{cases}$$

$$k \in \mathbb{Z}$$

Observe that if $x \neq k\pi$, we have:

$$\arcsin(\cos(x))\text{sgn}(\sin(x)) = \begin{cases} -x - \frac{\pi}{2} + 2k\pi, & x \in (-\pi + 2k\pi, 2k\pi) \\ -x + \frac{\pi}{2} + 2k\pi, & x \in (2k\pi, \pi + 2k\pi) \end{cases} = -x + \frac{\pi}{2} + k\pi, x \in (k\pi, (k+1)\pi) = \arctan(\cot(x))$$

where

$$\text{sgn}(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$$

By **Arctangent**, we get

$$\lfloor x \rfloor = x - \frac{1}{2} + \frac{1}{\pi} \arcsin(\cos(\pi x)) \text{sgn}(\sin(\pi x)), x \notin \mathbb{Z}$$

A continuous approximation of $\text{sgn}(x)$ which can also eliminate breakpoints is:

$$\text{sgn}(x) \approx \tanh(Mx) = 1 - \frac{2}{1 + e^{2Mx}}$$

where M is a large number

Combine them, we get:

$$\lfloor x \rfloor \approx x - \frac{1}{2} + \frac{1}{\pi} \phi(\pi x) \tanh(M \sin(x)) = x - 0.5 + \frac{\frac{\pi \cos(\pi x)}{\sqrt{1-\cos^2(\pi x)}} \left(1 - \frac{2}{1+e^{M \sin(\pi x)}}\right)}{4 + \sqrt{34 + \left(\frac{2\pi \cos(\pi x)}{\sqrt{1-\cos^2(\pi x)}}\right)^2}}$$

Now, without controlling ε and loss of precision, we get a stable approximation that can be combine with almost any function and works normally in Graphwar I

However, it's longer than the approximation in **3rd try**, and x appears 6 times, a higher frequency.

Notice that

$$\cot^2(\pi x) = \frac{1}{\tan^2(\pi x)} = \left(\frac{\cos(\pi x)}{\sqrt{1-\cos^2(\pi x)}} \right)^2$$

We get

$$x - 0.5 + \frac{\frac{\pi \cos(\pi x)}{\sqrt{1-\cos^2(\pi x)}} \left(1 - \frac{2}{1+e^{M \sin(\pi x)}}\right)}{4 + \sqrt{34 + \left(\frac{2\pi}{\tan(\pi x)}\right)^2}}$$

A natural idea is: What about $\frac{\cos(\pi x)}{\sqrt{1-\cos^2(\pi x)}}$?

It's clearly that:

$$\sin(x) = \begin{cases} \sqrt{1-\cos^2(x)}, & \sin(x) \geq 0 \\ -\sqrt{1-\cos^2(x)}, & \sin(x) \leq 0 \end{cases}$$

So we have

$$\sin(x) = \text{sgn}(\sin(x)) \sqrt{1-\cos^2(x)}$$

So

$$\cot(\pi x) = \frac{1}{\tan(\pi x)} = \frac{\cos(\pi x)}{\sin(\pi x)} = \frac{\cos(\pi x)}{\operatorname{sgn}(\sin(\pi x))\sqrt{1 - \cos^2(\pi x)}}$$

That is:

$$\frac{\cos(\pi x)}{\sqrt{1 - \cos^2(\pi x)}} = \frac{\operatorname{sgn}(\sin(\pi x))}{\tan(\pi x)} \approx \frac{1 - \frac{2}{1 + e^{M \sin(\pi x)}}}{\tan(\pi x)}$$

Finally we get:

$$\lfloor x \rfloor \approx x - 0.5 + \frac{\pi \left(1 - \frac{2}{1 + e^{M \sin(\pi x)}}\right)^2}{\tan(\pi x) \left(4 + \sqrt{34 + \left(\frac{2\pi}{\tan(\pi x)}\right)^2}\right)}$$

which is much shorter than the **3rd try** form and there are only 4 x 's in the formula smaller than the original 5.

Similarly, here is the expression that can be used in Graphwar: `x-0.5+(pi(1-2/(1+e^(500sin(pix))))^2)/(tan(pix)(4+sqrt(34+(2pi/tan(pix))^2)))`